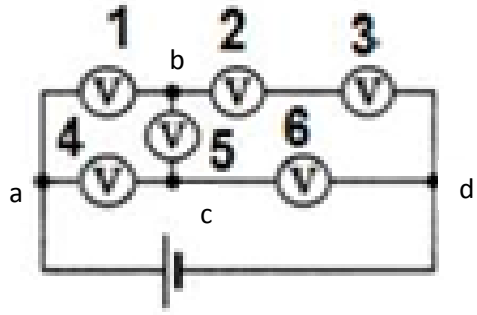
1. $\varepsilon = 6V$ $\frac{U_5}{U_3} = ?$

potensiallar tushuvi orqali kuchlanishlarni ifodalasak quyidagicha ko'rinishdabo'ladi

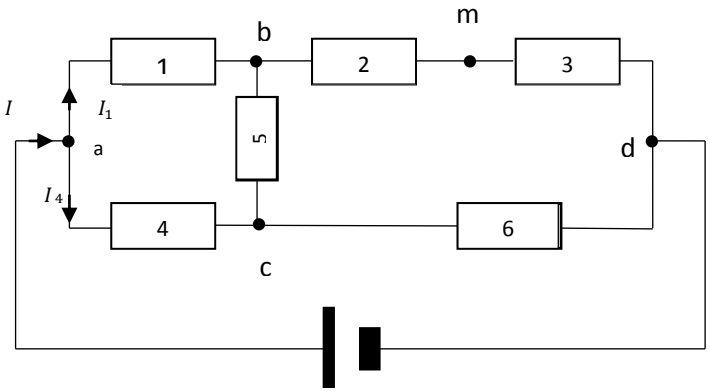


$\varphi_a - \varphi_b = U_1$

$\varphi_b - \varphi_d = U_2 + U_3$

$\varphi_a - \varphi_c = U_4$
$\varphi_c - \varphi_d = U_6$
$\varphi_b - \varphi_c = U_5$

Har bir voltmetr bir xil qarshilikka ega bo'lganligi uchun bu voltmetrlarni o'rniga rezistorlarni joylashtirib chiqamiz. Va chizmani boshqadan chizib olamiz



Krixgofning birinchi qonuniga ko'ra

$$I_1 = I_2 + I_5$$

$$I_6 = I_4 + I_5$$

Potensiallar ayirmasini qo'llaydigan bo'lsak quyidagi tenglamalar hosil bo'ladi. Barcha voltmetrlar bir xil R qarshilkka ega . $I_1 = \frac{\varphi_a - \varphi_b}{R}$; $I_2 = \frac{\varphi_b - \varphi_m}{R}$; $I_5 = \frac{\varphi_b - \varphi_c}{R}$; $I_6 = \frac{\varphi_c - \varphi_d}{R}$; $I_4 = \frac{\varphi_a - \varphi_c}{R}$

$$\frac{U_5}{U_3} = \frac{\varphi_b - \varphi_c}{\varphi_m - \varphi_d} = ?$$

$$\frac{\varphi_a - \varphi_b}{R} = \frac{\varphi_b - \varphi_m}{R} + \frac{\varphi_b - \varphi_c}{R}$$

$$\frac{\varphi_c - \varphi_d}{R} = \frac{\varphi_a - \varphi_c}{R} + \frac{\varphi_b - \varphi_c}{R}$$

Bu tenglamalardagi barcha R lar teng bo'lganligi uchun ularni qisqartirib yuboramiz

$$\varphi_a - \varphi_b = \varphi_b - \varphi_m + \varphi_b - \varphi_c$$

$$\varphi_c - \varphi_d = \varphi_a - \varphi_c + \varphi_b - \varphi_c$$

$$3\varphi_b = \varphi_a + \varphi_m + \varphi_c$$

$$3\varphi_c = \varphi_a + \varphi_b + \varphi_d$$

$$3(\varphi_b - \varphi_c) = \varphi_m + \varphi_c - \varphi_b - \varphi_d$$

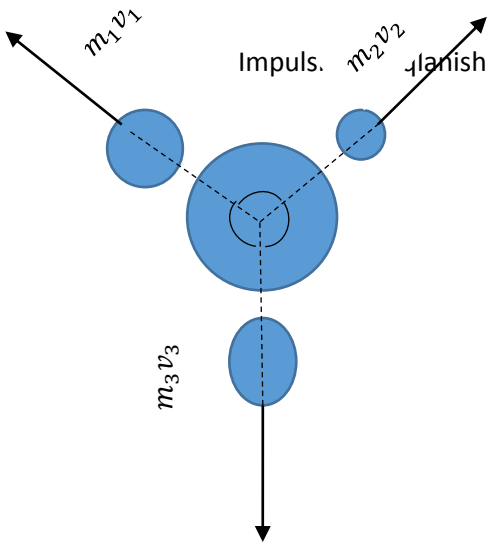
$$4(\varphi_b - \varphi_c) = \varphi_m - \varphi_d$$

$$\frac{U_5}{U_3} = \frac{\varphi_b - \varphi_c}{\varphi_m - \varphi_d} = \frac{1}{4}$$

2- masala

“Yadro” massalari m_1, m_2,m_3 bo’lgan uchta bo’lakka bo’lindi ular har biri o’zaro 120° dan burchak hosil qilib harakatlanib ketishdi. Ularning umumiy kinetic energiyasi E bo’lsa har bir bo’lakning tezligini toping

Dastlabgi holatda yadro tinch turibdi shuning uchun impulsi 0 ga teng bo’ladi.



Impuls. [illegible]lanish qonuniga ko’ra va har bir bo’laklar orasidagi burchak 120 ekanligini i

Demak :

$$m_1v_1 = m_2v_2 = m_3v_3$$

$$v_2 = \frac{m_1v_1}{m_2}$$

$$v_3 = \frac{m_1v_1}{m_3}$$

$$E = \frac{m_1{v_1}^2}{2} + \frac{m_2{v_2}^2}{2} + \frac{m_3{v_3}^2}{2}$$

Yuqoridagi tenglamalarni oxirgi tenglamaga qo’ysak

$$E = \frac{m_1{v_1}^2}{2} + \frac{m_2{v_2}^2}{2} + \frac{m_3{v_3}^2}{2} = \frac{m_1{v_1}^2}{2} + \frac{m_2{m_1}^2{v_1}^2}{2} + \frac{m_2{m_1}^2{v_1}^2}{2{m_2}^2} + \frac{m_3{m_1}^2{v_1}^2}{2{m_3}^2}$$

Bu tenglamadan

$$m_1{v_1}^2\left(1 + \frac{m_1}{m_2} + \frac{m_1}{m_3}\right) = 3E$$

Tenglama kelib chiqadi.

Bu tenglamadan

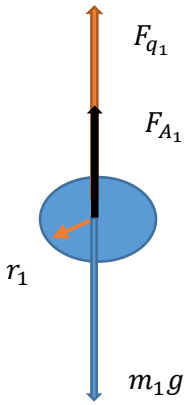
$$v_1 = \sqrt{\frac{2E * m_2m_3}{m_1m_2m_3 + m_2{m_1}^2 + m_3{m_1}^2}}$$

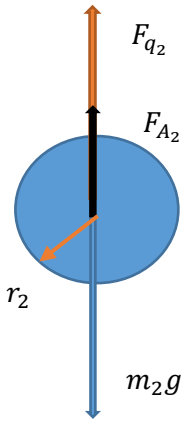
$$v_2 = \frac{m_1 v_1}{m_2} = \sqrt{\frac{2E * m_1 m_3}{m_1 m_2 m_3 + m_2 {m_1}^2 + m_3 {m_1}^2}}$$

$$v_3 = \frac{m_1 v_1}{m_3} = \sqrt{\frac{2E * m_1 m_2}{m_1 m_2 m_3 + m_2 {m_1}^2 + m_3 {m_1}^2}}$$

3- MASALA

Bir xil moddadan tayyorlangan ikkita sharcha havoda tusha boshlaydi. Sharchalar radiuslarining nisbati 4 ga teng. Katta sharchaning tushishdagi barqaror tezligi necha marta katta? Qarshilik kuchi sharchaning ko'ndalang kesim yuzasiga va uning tezligining kvadratiga proporsional. $\frac{r_2}{r_1} = 4$





Birinchi jism uchun tenglamani tuzsak

$$m_1 g = F_{A_1} + F_{q_1}$$

$$m_2 g = F_{A_2} + F_{q_2}$$

$$F_{A_1} = \rho_h V_1 g = \frac{4}{3}\rho_h {r_1}^3 g$$

$$F_{A_1} = \rho_h V_2 g = \frac{4}{3}\rho_h {r_2}^3 g$$

$$F_{q_1} = k{v_1}^2 \pi {r_1}^2 \qquad m_1 g = \rho V_1 g = \frac{4}{3}\rho {r_1}^3 g$$

$$F_{q_2} = k{v_2}^2 \pi {r_2}^2 \qquad m_2 g = \rho V_2 g = \frac{4}{3}\rho {r_1}^3 g$$

$$(k - qarshilik\ kuchining\ proporsionallik\ koeffitsiyenti\ )$$

$$m_1 g = F_{A_1} + F_{q_1} \mapsto \frac{4}{3}\rho {r_1}^3 g = \frac{4}{3}\rho_h {r_1}^3 g + k{v_1}^2 \pi {r_1}^2$$

$$m_2 g = F_{A_2} + F_{q_2} \mapsto \frac{4}{3}\rho {r_2}^3 g = \frac{4}{3}\rho_h {r_2}^3 g + k{v_1}^2 \pi {r_2}^2$$

bu tenglamalardan

$$\frac{4}{3}\rho r_1 g = \frac{4}{3}\rho_h r_1 g + k{v_1}^2 \pi$$

$$\frac{4}{3}\rho r_2 g = \frac{4}{3}\rho_h r_2 g + k{v_2}^2 \pi$$

Hosil bo'lgan bu tenglamalardan tezliklarni topsak

$$k{v_1}^2 \pi = \frac{4}{3}\rho_h r_1 g - \frac{4}{3}\rho r_1 g = \frac{4}{3} r_1 g(\rho_h - \rho)$$

$$k{v_2}^2 \pi = \frac{4}{3}\rho_h r_2 g - \frac{4}{3}\rho r_2 g = \frac{4}{3} r_2 g(\rho_h - \rho)$$

Bu ikki tenglamadan nisbat olsak

$$\frac{{v_1}^2}{{v_2}^2} = \frac{r_1}{r_2} = \frac{1}{4}$$

Hosil bo'ladi.

$$\frac{v_2}{v_1} = 2$$

kelib chiqadi.

4- MASALA

Raketa Yer sirtidan vertikal yuqoriga start oladi, va bunda tortish kuchi shunday o'zgartiriladiki, uning tezlanishi doimiy va 10 m/s^2 ga teng bo'lib qolaveradi. Raketaning tezligi 8 km/s ga teng bo'lguniga qadar uning massasi 10 % ga kamaydi. Ana shu paytgacha tortish kuchini necha foizga kamaytirish kerak? Yerning radiusi 6400 km, g = 10m/s^2.

$a = 10\frac{\text{m}_2}{\text{s}}$ $\qquad h = \frac{at^2}{2} = \frac{v^2}{2a} = 3200\ km$

$v = 8\frac{\text{km}}{\text{s}}$ $\qquad g_h = g\frac{r^2}{(r+h)^2} = g * \frac{4}{9}$

$\Delta m = 0{,}1m$ $\qquad F_1 = mg + ma = 20\ m$

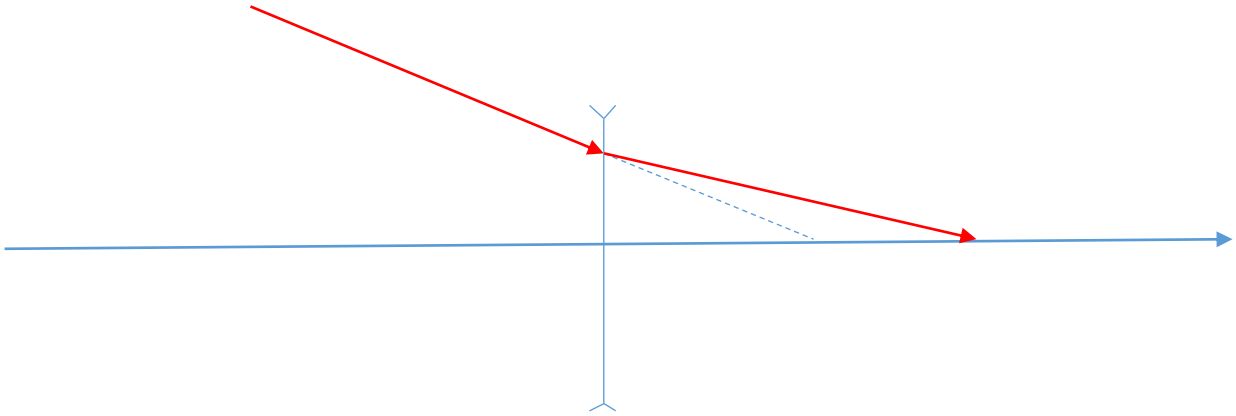
R=6400 km $\qquad F_2 = (m - \Delta m)g_h + (m - \Delta m)a = \frac{0{,}9m4g}{9} + 0{,}9ma = 13\ m$

$g = 10\frac{m_2}{s}$ $\qquad \frac{F_2}{F_1} = \frac{13m}{20m} = 0{,}65$

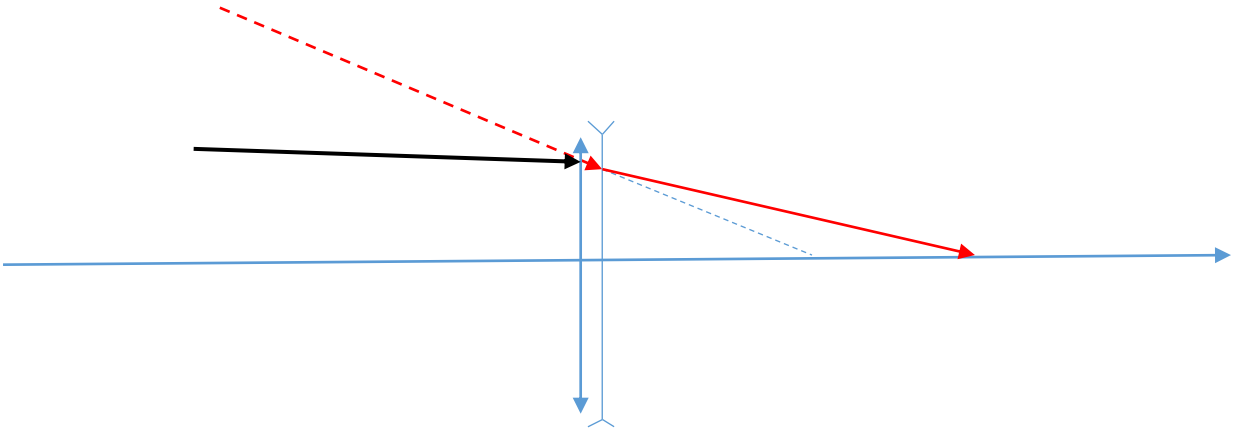
Demak 65 foizi qolgan bo'lishi kerak. Javob 35 % kamaytirish kerak.

5- masala

1. Yorug’lik (lazer) nuri fokus masofasi 13,5 sm bo’lgan sochuvchi linzadan sinib o’tib, linzadan 9 sm uzoqlikda bosh optik o’qni kesib o’tdi. Agar linza yo’q bo’lsa, nur bosh optik o’qni linza o’rnidan qanday uzoqlikda (sm) kesib o’tar edi?

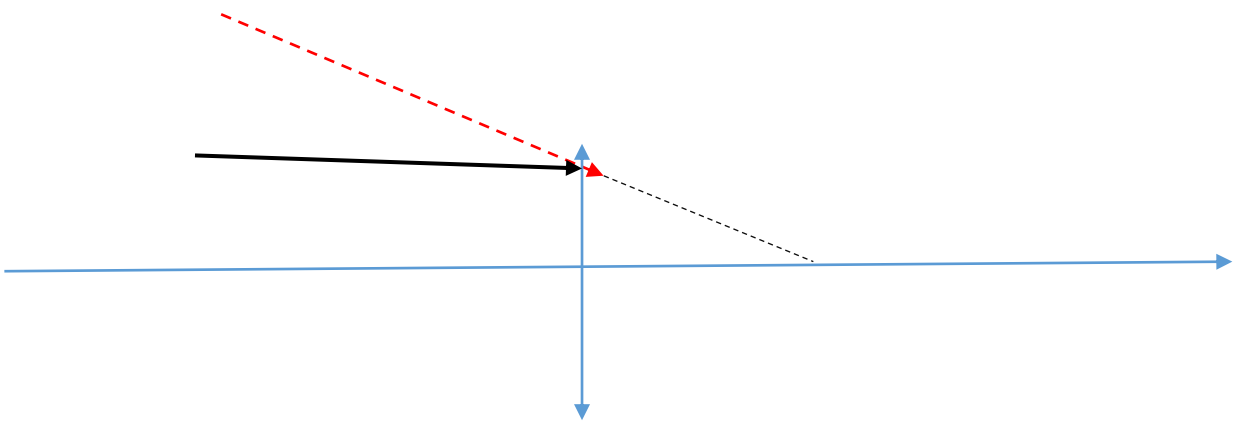


Masalaning chizmasi yuqoridagi ko’rinishda bo’ladi. Yani sochuvchi linzaga yig’iluvchi nurlar dastasi tushsagina u yig’ilishi mumkin. Bu yig’iluvchi nurlar dastasini bosh optik o’qqa parallel kelgan (chizmadagi qora rangli chiziq) nurlar dastasining oldiga yig’uvchi linza qo’yish orqali hosil qilish mumkin. Quyidagi chizma orqali ko’rishimiz mumkin.



Demak natijaviy optik kuch musbat bo’lishini bilib oldik chunki nurlar bitta nuqtada yig’iladi.

Natijaviy fokus masofa esa 9 sm ga teng bo’ladi. Agar bu sistemadan sochuvchi linzani olib tashlasak nurlar yig’uvchi linzaning fokusidan o’tadi. Yani yig’iluvchi nurlar sinmasdan yo’lida davom etar edi.



Demak bizga biz xayolan joylashtirgan yig’uvchi linzaning fokus masofasi kerak.

$$D_n = D_1 + D_2$$

$$D_n = \frac{100}{9}$$

$$D_1 = -\frac{1000}{135}$$

$$D_2 = D_n - D_1 = \frac{100}{9} + \frac{1000}{135} = \frac{2500}{135}$$

$$F_2 = \frac{1}{D_2} = \frac{135}{2500} = 0{,}054m = 5{,}4\ sm$$

Javob: 5,4 sm